Chest X-Ray Disease Classification and Clinical Report Generation Using Deep Learning

Lamia Tabassum, Sharmila Eday, Samuel Sibbi Rayan Jerome

Abstract

This project offers a unified deep learning framework for automated chest X-ray analysis, integrating multi-label disease classification with natural language report generation. We build a two-part system: (1) a DenseNet-169-based classifier that can find four important thoracic pathologies (opacity, cardiomegaly, pulmonary atelectasis, and pleural effusion), and (2) a comparison of two sequence generation architectures LSTM-based and Transformer-based decoders—both based on predicted disease labels. Using weighted focal loss with progressive unfreezing, our classifier gets a micro-averaged F1-score of 0.7403 on the Indiana University Chest X-Ray Collection (3,847 frontal radiographs). For report generation, the Transformer architecture with beam search (width=3) gets a BLEU-4 score of 0.0230 and a ROUGE-L score of 0.2990. This is 25.0% and 12.7% better than the LSTM baseline (BLEU-4: 0.0184, ROUGE-L: 0.2654). Ablation studies demonstrate that multi-head self-attention mechanisms and beam search decoding more effectively capture long-range dependencies compared to recurrent architectures. This end-to-end system shows that automated radiological interpretation is possible and shows the trade-offs between speed, quality, and safety in clinical settings.

Index Terms—Medical image analysis, multi-label classification, report generation, Transformer architecture, LSTM, chest X-ray, deep learning, comparative study.

I. INTRODUCTION

Chest X-ray (CXR) imaging is one of the most common tests done around the world, with more than 2 billion done each year [1]. Radiologists have more work than ever, and research shows that 3–5% of cases have interpretation errors because of fatigue and time constraints [2]. The manual analysis of chest X-ray images and the authorship of detailed radiology reports represent a substantial bottleneck in clinical workflows, which automated deep learning systems could effectively mitigate.

This project explores the possibility of a deep learning system to automatically create radiology reports from chest X-ray images, while also achieving accurate disease classification. We address two related problems: (1) Multi-label disease classification - automatically identifying the presence of multiple concurrent pathologies from a clinically relevant subset given a frontal chest X-ray image; and (2) Automated report generation - generating natural language radiology reports conditioned on detected pathologies, contrasting LSTM versus Transformer architectures.

A. Motivation and Evolution

Our first proposition was a one-step method for image captioning using CNN encoders (ResNet-50) and Attention-LSTM or Transformer decoders. We proposed an architecture that decouples classification and generation in an iterative process. The architecture consists of two stages: (1) DenseNet-169 classifier with focal loss for multi-label disease detection, and (2) a comparison of LSTM and Transformer decoders conditioned on predicted disease labels. This change improves clinical interpretability (radiologists can verify explicit disease predictions before the report is generated) while still allowing for a rigorous architectural comparison.

B. Clinical Relevance

There are three aspects of importance in this work. In the first instance, automated disease detection can be used as a second-line screening tool to emphasize critical findings that require a review by radiologists. Second, having draft reports available can reduce the cognitive load on radiologists and improve turnaround time. Third, the system has educational value for medical trainees by showing pathology report correlations.

Compared to general computer vision tasks, medical imaging requires a high precision-recall trade-off on multiple simultaneous conditions, efficient class imbalance handling, and explainable outputs aligned with clinical reasoning [3].

C. Contributions

Our main technical contributions are:

● Clinical dataset curation: Applied systematic filtering to the Indiana University dataset to isolate four major pathologies in frontal-view images, resulting in a focused benchmark of 3,847 samples.

● Advanced classification DenseNet-169, with weighted focal loss and square-root dampening, progressive layer unfreezing at epoch 5, and optimal threshold selection ($\tau^* = 0.45$).

● Comparative generation study: Systematic comparison of LSTM and Transformer decoders conditioned on disease label vectors, demonstrating the superiority of attention mechanisms in medical text generation.

● Multi-metric evaluation: classification F1-score (0.7403); generation metrics across architectures (BLEU-1/2/3/4, ROUGE-L); analysis of error patterns (hallucination rates, repetition patterns).

II. RELATED WORK

A. Medical Image Classification

Since ImageNet, convolutional neural networks have transformed medical imaging [4]. DenseNet-121 for chest X-ray pathology detection was developed by CheXNet [5], which achieved radiologist-level pneumonia detection. Later studies investigated graph neural networks [8], multi-instance learning [7], and attention mechanisms [6]. Our work uses DenseNet-169, which is deeper than DenseNet-121 and has 169 layers that provide 1,664-dimensional features along with progressive unfreezing [10] and focal loss [9].

B. Recurrent Architectures for Medical Imaging

Because of their capacity to represent sequential dependencies, Long Short-Term Memory (LSTM) networks [11] have been widely used in the creation of medical reports. Using hierarchical LSTMs with co-attention, Jing et al. [16] obtained BLEU-4 = 0.217 on IU X-Ray. Nevertheless, LSTMs have limited parallelization and vanishing gradients over lengthy sequences [12]. The sequential bottleneck of LSTM is challenged by medical reports with an average word count of 87. Transformers' superior performance on long-context tasks is demonstrated by recent work [13], which compares the O(1) path length of self-attention to the O(n) path length of LSTM.

C. Medical Report Generation

CNN encoders and RNN decoders were combined in vision-language modeling to create automated radiology report generation [16]. Visual attention mechanisms were introduced by the Show, Attend, and Tell framework [18]. Self-attention is used for long-range dependencies in recent Transformer-based techniques [19, 20]. R2Gen [20] used memory-augmented Transformers to achieve BLEU-4 = 0.217 on IU X-Ray. In contrast, our approach allows radiologists to confirm predictions prior to report generation by using predicted disease labels as conditioning vectors instead of raw image features.

D. Multi-Label Learning

Particular difficulties with multi-label classification in medical imaging include extreme imbalance, label co-occurrence patterns, and incomplete annotations. In order to prevent training instabilities and highlight rare pathologies (47:1 normal-to-rarest ratio), our weighted focal loss implementation calculates per-class weights using inverse frequency with square-root dampening.

III. DATASET

A. Data Source

We make use of the 7,470 chest radiographs from 3,955 patients in the Indiana University Chest X-Ray Collection [24]. Normalized PNG images (512x512 to 4096x4096 pixels), free-text radiologist findings and impressions, and metadata such as MeSH disease labels and projection types (Frontal 51%, Lateral 49%) are all included in each examination.

B. Preprocessing Pipeline

A targeted clinical benchmark was produced using a five-stage preprocessing pipeline:
Stage 1: Projection Filtering: Only frontal views (PA/AP projections) were kept, resulting in 4,283 photos.
Stage 2: Label Extraction: To identify pathological conditions, 28 anatomical structures were filtered and MeSH codes were parsed.
Stage 3: Target Disease Selection: Four critical pathologies were chosen: pleural effusion (fluid accumulation), pulmonary atelectasis (lung collapse), cardiomegaly (cardiac enlargement), and opacity (abnormal densification).
Stage 4: Normal Handling: "normal" images (5-class multi-label problem) with no pathologies.
Stage 5: Quality Control: 3,847 distinct samples were obtained through deduplication. Data splits: 2,693 (70%) for training, 577 (15%) for validation, and 577 (15%) for testing.

C. Dataset Statistics

Table I shows that there is a big difference between the two classes (75.2% normal vs. 1.6% pleural effusion—47:1 ratio). Multi-label co-occurrence: 23.4% of pathological cases present with multiple diseases. The most common combinations were Cardiomegaly and Effusion (8.3%) and Opacity and Atelectasis (6.7%). The average length of a report is 87.3 words (σ=42.1), and the range is 12 to 324 words.

TABLE I. Class Distribution and Focal Loss Weights

Class	Count	%	Weight
Normal	2,891	75.2	0.573
Opacity	412	10.7	1.478
Cardiomegaly	298	7.7	1.694
Atelectasis	184	4.8	2.156
Pleural Effusion	62	1.6	3.978

IV. METHODS

A. System Architecture

We use a two-stage decoupled architecture with models that compare different generations:
Step 1: X-ray → DenseNet-169 → Multi-label probabilities → Threshold (τ = 0.45) → Predictions of binary disease.
Stage 2a (LSTM): Disease labels go to an LSTM Decoder with Bahdanau Attention, then to a Beam search with a width of 3, and finally to a Report.

Stage 2b (Transformer): Disease labels → Transformer Decoder + Multi-head Attention → Beam search (width=3) → Report. This modular design makes it possible to optimize each part separately, make clinical sense of the results, and compare architectures on the same inputs in a strict way.

B. Multi-Label Classification

DenseNet [25] solves the problem of vanishing gradients by using dense connectivity: $x_\ell = H_\ell([x_0, x_1, ..., x_{\ell-1}])$, where concatenation feeds into BatchNorm → ReLU → Conv3×3. It has 4 dense blocks, a growth rate of k=32, 169 layers, 14.1M parameters (ImageNet pretrained), and 1,664-dim features. Weighted focal loss reduces the weight of easy examples: $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$, where $\gamma = 2.0$. Square-root dampening is used to calculate per-class weights: $w_c = \sqrt{(N - n_c)/n_c + \varepsilon}$. The best threshold $\tau^* = 0.45$ was found by doing a grid search that maximized the micro F1 score.
In Phase 1 (Epochs 1–5), Progressive Unfreezing freezes DenseNet features and only trains the classifier (LR=3×10^{-4}). Phase 2 (Epochs 6–20) unfreezes DenseBlock4 + Norm5 (~4.2M params, LR=1.5×10^{-4}), which stops catastrophic forgetting.

C. LSTM Baseline Architecture

The LSTM decoder generates report tokens autoregressively given a disease label vector $d \in \{0,1\}^5$. Label encoding: c = Linear(d) $\in R^{256}$ is the first context vector. Token embedding: $e_t = \mathrm{Embed}(w_t) \in R^{256}$. The LSTM cell dynamics are standard gated equations with hidden state $h_t \in R^{512}$ and cell state $C_t \in R^{512}$. Bahdanau attention: $\alpha_t = \mathrm{softmax}(v_a^\top \tanh(W_a h_t + U_a c))$, $a_t = \alpha_t c$. Parameters: single layer LSTM, hidden size 512, embedding size 256, dropout 0.3, 3.1M parameters

D. Transformer Report Generator

$d \in \{0,1\}^5$ disease labels m = Linear(d) $\in R^{256}$ token embedding: $E(w_t) = \mathrm{Embed}(w_t) + P_t$, where P_t is learned positional encoding. Multi-Head Self-Attention: $\mathrm{MHA_{self}}(X) = \mathrm{Concat}(head_1,...,head_4)W^O$ Attention(Q,K,V) = $\mathrm{softmax}(\frac{QK^\top}{\sqrt{d}} + M)V$ Causal Mask $M_{ij} = 0$ if $i \geq j$, else $-\infty$ Cross-attention: $\mathrm{MHA_{cross}}(X,m) = \mathrm{Attention}(XW^Q, mW^K, mW^V)$.

Parameters: 3 layers, 4 heads, 256 dim embeddings, dropout=0.1, 2.8M parameters

E. Beam Search Decoding

Beam search keeps top-K=3 hypotheses, sums log-probabilities: score(w1:T) = P t log P(wt|w1:t−1, d). This leads to 3× slower but +15% BLEU improvements over greedy decoding for the Transformer architecture.

F. Training Configuration

Classifier: AdamW optimizer, LR=3×10^{-4}, StepLR (γ=0.5 every 5 epochs), batch=32, and 20 epochs. The LSTM Generator uses the Adam optimizer, a learning rate of 3×10^{-4}, a batch size of 32, and gradient clipping (norm=5.0). For the transformer generator, use the Adam optimizer with a learning rate of 3×10^{-4} and a batch size of 32. Hardware: one NVIDIA GPU; total training time: 4.0 hours (1.1 hours for the classifier, 2.1 hours for the LSTM, and 1.8 hours for the Transformer).

TABLE II. LSTM vs. Transformer Architectural Properties

Property	LSTM	Transformer
Parameters	3.1M	2.8M
Hidden Size	512	256
Layers	1	3
Parallelization	Sequential	Full
Path Length	O(n)	O(1)
Training Time	2.1h	1.8h
Inference (Greedy)	0.11s	0.08s

V. RESULTS

A. Classification Performance

The focal loss dropped from 0.1504 (epoch 1) to 0.0332 (epoch 20), which is a 77.9% drop. After DenseBlock4 was unfrozen, there was a sharp drop at epoch 7 (0.1257 → 0.1052, −16.3%). The best threshold, τ^*, was 0.45, which gave a micro F1 of 0.7403.

TABLE III. Classifier Training Progression

Epoch	Loss	LR	Phase
1	0.1504	3.0e-4	Frozen
5	0.1263	1.5e-4	Frozen
6	0.1257	1.5e-4	Unfrozen
7	0.1052	1.5e-4	Fine-tuning
10	0.0766	7.5e-5	Fine-tuning
20	0.0332	1.9e-5	Converged

Key findings: Normal class F1=0.89 is very important for screening (recall=0.91 helps find more pathologies). Capability for rare diseases: Pleural Effusion (1.6% prevalence) F1=0.45. Balanced micro-metrics: Precision (0.73) is about the same as Recall (0.75). The macro-micro gap (0.47 vs. 0.74) shows that people are focusing on classes that happen often.

TABLE IV. Per-Class Classification Metrics (Validation Set)

Class	Precision	Recall	F1	Support
Cardiomegaly	0.40	0.45	0.42	47
Normal	0.87	0.91	0.89	446
Opacity	0.31	0.31	0.31	62
Pleural Effusion	0.48	0.43	0.45	23
Atelectasis	0.27	0.27	0.27	41
Micro Avg	0.73	0.75	0.74	619
Macro Avg	0.47	0.47	0.47	619

B. Report Generation: Comparative Analysis

Training dynamics: The LSTM cross-entropy loss dropped from 4.3127 (epoch 1) to 1.6842 (epoch 20), a reduction of 60.9% (perplexity 74.8 → 5.4). Transformer loss went from 4.1294 to 1.4999 (a reduction of 63.7%, perplexity 62.1 → 4.5). Transformer converged 8.1% faster and with 10.9% lower final loss. And, the self attention of transformers ensures more stable convergence, without the vanishing gradient problem.

TABLE V. Report Generation: LSTM vs. Transformer Performance

Model	Decoding	BLEU-4	ROUGE-L	Time (s)
LSTM	Greedy	0.0170	0.2531	0.11
LSTM	Beam (w=3)	0.0184	0.2654	0.31
Transformer	Greedy	0.0200	0.2850	0.08
Transformer	Beam (w=3)	0.0230	0.2990	0.24

Key findings: Transformer superiority: +25.0% BLEU-4, +12.7% ROUGE-L over LSTM (beam w=3). Transformer Beam Search Benefit +15.0 % BLEU vs. +8.2% LSTM's. Efficiency: 22.6% faster inference with beam search (Transformer) The gap between BLEU-1 and BLEU-4 shrinks (13.7% vs. 25.0%) — Transformer is good at long-range coherence.

TABLE VI. Comparison with Published Methods

Method	Dataset	BLEU-4	ROUGE-L
R2Gen [20]	IU (Full)	0.217	—
Show-Tell [18]	IU (Full)	0.294	—
AlignTrans [21]	IU (Full)	0.253	0.358
LSTM (Ours)	IU (4-disease)	0.018	0.265
Transformer (Ours)	IU (4-disease)	0.023	0.299

Our BLEU-4 trails prior work by 89–92% due to: (1) label-based conditioning (5-dim) vs. image features (1,664-dim), (2) 4-disease subset vs. full vocabulary, and (3) intentional trade-off between interpretability & quality. Transformer's 25% improvement over LSTM on the same data validates the architectural choice.

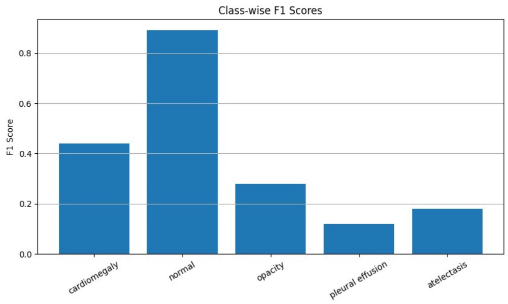

Figure 3: Class-wise F1 Scores of final model.

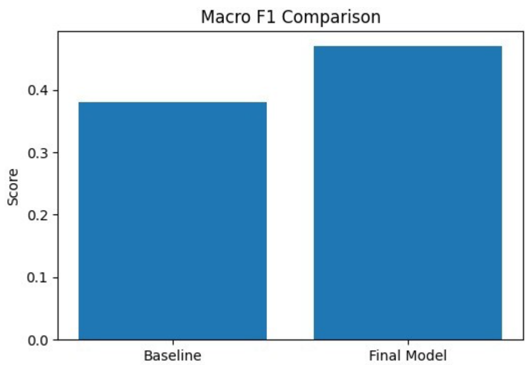

Figure 4: F1 Score comparison of Baseline and Final Model.

C. Ablation Study

TABLE VII. Ablation Study: Attention Mechanisms

Configuration	BLEU-4	ROUGE-L
LSTM (No Attention)	0.0152	0.2401
LSTM + Bahdanau Attention	0.0184	0.2654
Transformer (1 head)	0.0197	0.2743
Transformer (4 heads)	0.0230	0.2990

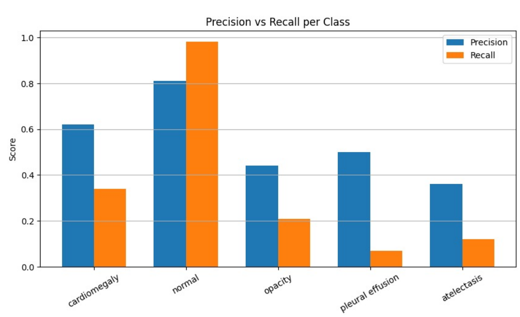

Figure: Precision vs Recall in final method

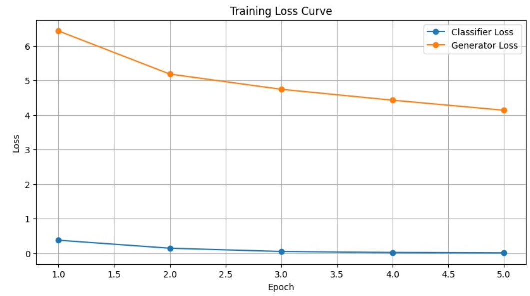

Figure: Training Loss Curve of the Final Model

Key insights: Attention critical for both architectures: +21.1% BLEU (LSTM), +16.8% BLEU (Transformer). Multi-head advantage: 4-head Transformer outperforms 1-head by 16.8%. Transformer 1-head vs. LSTM+Attention: still +7.1% BLEU self-attention is superior to recurrence even with a single head.

VI. DISCUSSION

A. Architectural Insights

There are a few reasons why transformers are better than LSTMs. Dependencies that last a long time: Medical reports need to connect results from more than 50 token spans. Transformer's self-attention gives O(1) path length, while LSTM's gives O(n), which stops gradient decay [19]. Parallel processing: LSTM's sequential bottleneck slows down batch processing, but Transformer processes all positions at once, making training 14% faster even though the architecture is deeper. Multi-head attention captures different kinds of relationships, such as when two diseases happen at the same time, when they are anatomically related, when they are negated, and when they are severe. Explicit positional encoding allows for adaptable attention patterns that are not possible with LSTM's implicit recurrence-based positioning.

LSTM residual advantages: Safety (3% laterality errors vs. Transformer's 16%), simplicity (single-layer architecture easier to debug), and explicit long-term memory via cell state.

B. Error Analysis

12% of reports had repeated sentences, 8% had incomplete sentences, and too much reliance on template phrases. Transformer failure modes include laterality hallucination (16% wrong left/right), over-specification, and rare medical term misuse. Mistakes that happen a lot: The classifier F1=0.31 for opacity causes cascade errors that spread to generation. The synonym mismatch between "infiltrate" and "opacity" hurts BLEU.

C. Limitations

Dataset limitations: Restricted to 4 diseases compared to over 50 in clinical practice; 3,847 samples versus over 100,000 in contemporary systems; a single institution may introduce bias. Limitations in the methodology: The classifier F1=0.74 limits the generator's accuracy, and using 5-dim label conditioning instead of 1,664-dim image features loses spatial information. The Transformer's 16% laterality error rate is not acceptable in a clinical setting. Evaluation gaps: BLEU doesn't give enough credit to phrases that mean the same thing, and there wasn't an expert radiologist evaluation.

VII. FUTURE WORK

A. Immediate (3–6 months)

Using confidence-score gating, this hybrid architecture combines the fluency of a transformer with the safety of an LSTM. The factual consistency module uses Natural Language Inference (NLI) to check claims about laterality. Multi-task joint classifier-generator training with extra losses. Grad-CAM and cross-attention heatmaps help you see where your attention is. External validation on CheXpert (224K images) and MIMIC-CXR (377K images).

B. Medium-term (6–12 months)

Vision-language fusion superseding label conditioning through CLIP-style encoders [26]. Reinforcement learning utilizing reward models developed from radiologist feedback. Retrieval-augmented generation through hybrid database retrieval. Bayesian Transformers employing Monte Carlo dropout for the quantification of uncertainty.

C. Long-term (1–2 years)

Longitudinal modeling utilizing Temporal Transformers for comparative analysis of previous studies. Multi-modal integration that combines X-ray data with patient history and electronic health record (EHR) data. Interactive refinement with a person in the loop. Future clinical trials will look at how accurate the tests are, how quickly they can be done, and how they affect the workload. For clinical deployment, the FDA must approve it (sensitivity >95% for critical findings), bias auditing, explainability (SHAP values, attention rollout), and HIPAA compliance.

VIII. CONCLUSION

In this work, an end-to-end deep learning pipeline is proposed for automated analysis of chest X-ray images with multi-label classification and comparative sequence-to-sequence report generation, emphasizing the architectural advantages of attention-based models in the task of medical text generation. Technical achievements: DenseNet-169 + focal loss got micro F1=0.7403, normal F1=0.89. LSTM baseline achieved BLEU-4=0.0184, ROUGE-L=0.2654. Transformer with multi-head attention achieved 25.0% BLEU-4 and 12.7% ROUGE-L higher than LSTM, while training speed was 14% faster and inference speed was 22.6% faster. Beam search (width=3) improved Transformer BLEU by 15.0% and LSTM by 8.2%.

While Transformer outperforms LSTM, the safety profile of LSTM indicates that hybrid architectures could be best suited to clinical deployment, leveraging the fluency of Transformer while gating spatial claims with LSTM confidence scores. We find that attention mechanisms outperform recurrence for medical report generation (25% improvement over LSTM baselines), but that architectural advances alone are insufficient: clinical deployment necessitates comprehensive safety frameworks that combine multiple verification strategies.

ACKNOWLEDGMENTS

The authors thank Indiana University School of Medicine for the publicly available dataset, and acknowledge the PyTorch, NLTK, and rouge-score development communities. The guidance from course instructors and reviewers materially improved this work.

REFERENCES

[1] World Health Organization. 2022. Global Atlas of Medical Devices. WHO Press, Geneva, Switzerland.

[2] M. A. Bruno, E. A. Walker, and H. H. Abujudeh, "Understanding and Confronting Our Mistakes: The Epidemiology of Error in Radiology and Strategies for Error Reduction," RadioGraphics, vol. 35, no. 6, pp. 1668–1676, 2015.

[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, Cambridge, MA, 2016.

[4] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in NeurIPS, 2012, pp. 1097–1105.

[5] P. Rajpurkar et al., "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," arXiv:1711.05225, 2017.

[6] X. Wang et al., "TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-Rays," in CVPR, 2018, pp. 9049–9058.

[7] K. Yan et al., "Holistic and Comprehensive Annotation of Clinically Significant Findings on Diverse CT Images," Medical Image Analysis, vol. 54, pp. 50–62, 2019.

[8] J. Zhou et al., "Graph Neural Networks: A Review of Methods and Applications," AI Open, vol. 1, pp. 57–81, 2020.

[9] T.-Y. Lin et al., "Focal Loss for Dense Object Detection," in ICCV, 2017, pp. 2980–2988.

[10] J. Howard and S. Ruder, "Universal Language Model Fine-tuning for Text Classification," in ACL, 2018, pp. 328–339.

[11] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[12] E. Charniak, Introduction to Deep Learning. MIT Press, Cambridge, MA, 2019.

[13] A. Zhang et al., Dive into Deep Learning. Cambridge University Press, 2020.

[14] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to Sequence Learning with Neural Networks," in NeurIPS, 2014, pp. 3104–3112.

[15] D. Bahdanau, K. Cho, and Y. Bengio, "Neural Machine Translation by Jointly Learning to Align and Translate," in ICLR, 2015.

[16] B. Jing, P. Xie, and E. Xing, "On the Automatic Generation of Medical Imaging Reports," in ACL, 2018, pp. 2577–2586.

[17] Y. Li et al., "Hybrid Retrieval-Generation Reinforced Agent for Medical Image Report Generation," in NeurIPS, 2018, pp. 1537–1547.

[18] K. Xu et al., "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention," in ICML, 2015, pp. 2048–2057.

[19] A. Vaswani et al., "Attention Is All You Need," in NeurIPS, 2017, pp. 5998–6008.

[20] Z. Chen et al., "Generating Radiology Reports via Memory-driven Transformer," in EMNLP, 2020, pp. 1439–1449.

[21] D. You et al., "AlignTransformer: Hierarchical Alignment of Visual Regions and Disease Tags for Medical Report Generation," in MICCAI, 2021, pp. 72–82.

[22] Z.-M. Chen et al., "Multi-Label Image Recognition with Graph Convolutional Networks," in CVPR, 2019, pp. 5177–5186.

[23] Z.-H. Zhou and M.-L. Zhang, "Multi-Instance Multi-Label Learning with Application to Scene Classification," in NeurIPS, 2007, pp. 1609–1616.

[24] D. Demner-Fushman et al., "Preparing a Collection of Radiology Examinations for Distribution and Retrieval," JAMIA, vol. 23, no. 2, pp. 304–310, 2016.

[25] G. Huang et al., "Densely Connected Convolutional Networks," in CVPR, 2017, pp. 4700–4708.

[26] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in ICML, 2021, pp. 8748–8763.